DE–4151 **11**

DISTANCE EDUCATION

M.Sc. (Botany) DEGREE EXAMINATION, MAY 2012.

Specialization—Plant Biotechnology

PLANT DIVERSITY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks

(5 × 20 = 100)

1. (a) What are the various habitats in which the Rhodophyta are found? (7)

   (b) What are thermophillic fungi? Where are they found? (7)

   (c) Describe the life-cycle of a moss. (6)

2. (a) Give an account of fossil bryophyte. (6)

   (b) Describe heterospory and seed habit in Pteridophyta. (7)

   (c) Write notes on conifers. (7)

3. (a) Describe the diversity in the vegetative structure among the algae. (7)

   (b) Describe the patterns of life cycle found in fungi. (6)

   (c) What are the salient features of Gnetum? (7)

4. (a) Give an outline classification of the Bryophyta. (7)

   (b) What is the economic importance of the Pteridophyta. (6)

   (c) Compare the distribution of the red and brown algae. (7)

5. (a) Compare and contrast between a moss and a liverwort. (7)

(b) Describe the Phenomena-apogamy and apospory in the pteridophyta. (7)

(c) Give an account of the geographic distribution of the gynmosperm. (6)

6. (a) Describe the thallus organization in the fungi. (6)

(b) What do you understand of the pigment constitution in the various groups of algae? (7)

(c) Give a brief account of the evolution of stele in the pteridophyta. (7)

7. (a) What are the bryophyta limited to moist areas? (6)

(b) Discuss fossil pteridophytes. (7)

(c) Enumerate Ginko biloba (7)

8. (a) Describe the spore dispersal mechanisms as found in the fungi. (7)

(b) Describe the life-history of Polysiphonia . (6)

(c) What are the various methods of vegetative reproduction found in the bryophyta? (7)

DE–4152 **12**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, MAY 2012.

Botany (Specialization Plant Biotechnology)

PLANT TAXONOMY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carry 20 marks.

$(5 \times 20 = 100)$

1. (a) What are the scope and objectives of plant taxonomy. (6)

   (b) Explain evolutionary species concept. (7)

   (c) What is a binomial system of nomenclature? Explain in detail. (7)

2. (a) Comment on decandoll's contribution to taxonomy. (6)

   (b) What are the merits and demerits of Benthan and Hooker's system of classification. (7)

   (c) Enumerate the principle laid out for Hutchinsons system of classification. (7)

3. (a) Write about the Engler and Prantl classification. (7)

   (b) Enumerate the merits and demerits of Takhtejans classification. (6)

   (c) Comment on the bio systematic categories. (7)

4. (a) Write short notes on common names and scientific names. (6)

(b) Define OTU and attributes. (7)

(c) What is a 'Flora'? Comment on the florisitic studies in Tamilnadu. (7)

5. (a) Describe serotaxonomy and its importance. (7)

(b) Explain the principles of ICBN. (6)

(c) Write about authoricitation. (7)

6. (a) Explain the rule of type concept. Point out any five types with example. (7)

(b) Briefly explain effective and valid publication. (7)

(c) Explain nomina conservanda. (6)

7. (a) What do you know about principles of priority. (6)

(b) Narrate the salient features of Draft Biocode. (7)

(c) Ranunclaceae is considered as primitive family. Justify. (7)

8. (a) What are the salient features of the family Rhamnae? Mention three economically important plants. (7)

(b) Assign viscum to its family. Mentio its dianostic features. (6)

(c) Orchidaceal is a very highly advanced family substantiate. (7)

DE–4153 **13**

DISTANCE EDUCATION

M.Sc. Botany (Specialization – Plant Biotechnology)
DEGREE EXAMINATION, MAY 2012.

ANATOMY AND EMBRYOLOGY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carry 20 marks.

(5 × 20 = 100)

1. (a) Write the functional properties of cellular organelles. (8)

   (b) Draw a detailed account of Parenchyma, Sclerenchyma and colenchyma. (6)

   (c) Enumerate the structural diversity and phylogenetic specialization of phloem. (6)

2. (a) Describe the vascular differentiation in the primary body of stem, root and leaf. (12)

   (b) Write short notes on root stem transition. (3)

   (c) Comment on lignin, and pectin. (5)

3. (a) Explain compression wood and reaction wood. (8)

   (b) Define natural defects of wood. (5)

   (c) Describe the molecular aspects of wood differentiation. (7)

4. (a) Write the stages of anther development with suitable diagrams. (8)

(b) Write notes on pollen morphology. (5)

(c) What is vegetative reproduction, explain. (7)

5. (a) Describe molecular aspects of higher plant reproduction. (10)

(b) Write about exploitation of polyembryony and apomixes in plant improvement programmes. (10)

6. (a) Write short notes on types of endosperm. (6)

(b) Write an essay about the structure and classification of woods. (10)

(c) Comment on Gametogenesis. (4)

7. (a) Explain the process of utilization of nutrition of embryo sac. (7)

(b) Describe chemical and mechanical properties of wood. (10)

(c) Short notes on Apomixis. (3)

8. (a) Draw detail study of cell structure with aid of Light and Electron microscope. (12)

(b) Write the significance of plant transfer cells. (5)

(c) Comment on Non-Storied vascular cambium. (3)

DE–4154 **14**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2012.

PLANT TISSUE CULTURE

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carry 20 marks.

($5 \times 20 = 100$)

1. (a) What are the aseptic condition should follow on plant tissue culture. (3)

   (b) Explain the methodology of Production of pathogen free plants. (10)

   (c) Write an essay on Generation of Callus and suspension culture from monocot and dicot plants. (7)

2. (a) Explain in detail on preparation of solid and liquid culture. (6)

   (b) Describe various of stages and types of explants for commercial propagation. (10)

   (c) Comment on floriculture. (4)

3. (a) Narrate about somatic embryogenesis. (6)

   (b) Examine the role of hormones in plant regeneration. (7)

   (c) Mention the control criteria of organogenesis in detail. (7)

4. (a) Describe the significance of haploid culture. (8)

(b) Write experimental explanation of in vitro pollination. (8)

(c) How would you conserve forest through plant tissue culture? (4)

5. Explain in detail on

(a) Macro nutrients, Micro nutrients and Inorganic nutrients for plant tissue culture. (10)

(b) Write the principles and protocol of protoplast isolation. (8)

(c) What is LUX meter. (2)

6. (a) Explain embryo rescue and their applications. (10)

(b) Write the impact of Transgenic plant cultivation and environment. (6)

(c) Cryopreservation. (4)

7. (a) Bioreactor. (8)

(b) Describe Somatic hybridization. (8)

(c) Hypochiorite solution. (4)

8. (a) Somoclonal variation-significance. (6)

(b) Write an essay on Germplasm conservation. (8)

(c) Essay about *In vitro* production of secondary plant products. (6)

DE–4155 **15**

DISTANCE EDUCATION

M.Sc.(Botany) (Specialization Plant Bio-technology) DEGREE EXAMINATION, MAY 2012.

PLANT MOLECULAR BIOLOGY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carry 20 marks.

($5 \times 20 = 100$)

1. (a) Write an essay about genomic organization of Plant. (10)

   (b) Comment on Agrobacterium tumefaciens. (6)

   (c) Distinguish between Chloroplast genes and mitochondrial genes. (4)

2. (a) Give a detailed account of plant hormones. (8)

   (b) Short notes on transposons. (4)

   (c) Explain regulation of gene expression in plants. (8)

3. (a) Examine how the Pea plant getting growth stimulator from the source of atmosphere. (10)

   (b) Write short notes on Binary vectors. (4)

   (c) What are the aspects involved T-DNA transfer in eukaryotes? (6)

4. (a) Narrate essay on Plant stress response. (12)

(b) Name the few novel Plant products through DNA manipulation. (3)

(c) Describe any one of the physical gene transfer method for plant. (5)

5. (a) What are molecular pharming and how it done? (7)

(b) Enumerate the process environment cleaning by plants with suitable examples. (7)

(c) Short notes on translation process. (6)

6. (a) Seed storage proteins – classes, properties and function. (8)

(b) Give the exact details about Agro infection. (7)

(c) Impacts of green revolution. (5)

7. Explain briefly with diagrams.

(a) Gene tagging (4)

(b) Genome Mapping (6)

(c) Cloning. (10)

8. (a) How the crown gall artificially created by researchers? Perfect diagrams needed. (10)

(b) Write briefly about plant cellular organelle genomes. (7)

(c) What is transgenic ? Explain. (3)

DE–4156

21

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, MAY 2012.

Botany (Specialization Plant Biotechnology)

PLANT PHYSIOLOGY AND BIOCHEMISTRY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the various phenomena found in plants

   (i) Diffusion

   (ii) Osmosis

   (iii) Plasmolysis. (6)

   (b) Explain the process of absorption of water in land plants. (7)

   (c) Give a brief account of the mechanism of stomatal movement. (7)

2. (a) Give a detailed notes on ultra structure of chloroplast. (6)

   (b) Difference between C3 and C4 plants. (7)

   (c) How does photorespiration differ from dark respiration? Discuss critically. (7)

3. (a) Explain the process of glycolysis. (7)

(b) Discuss briefly about the biological nitrogen fixation. (6)

(c) Comment on electron transport in mitochondria. (7)

4. (a) Proteins are called as biological primers. Explain.(6)

(b) What are carbohydrates and write about various forms of carbohydrates found in nature. (8)

(c) Comment on stereoisomer, enantiomers and epimers. (6)

5. (a) What are the three common type of membrane lipids. Comment on it. (6)

(b) Describe the α-helix and β-sheet of protein molecule. (7)

(c) Explain the biosynthesis of fatty acids. (7)

6. (a) What do you mean by allosteric mechanism of enzymes. (7)

(b) Explain Michaelis — Menton equation and its significance. (7)

(c) Write about the classification of lipids. (6)

7. (a) Differentiate active absorption and passive absorption. (7)

(b) Transpiration is often described as necessary event. Justify this statement. (7)

(c) Write short notes on :

(i) imbibitions

(ii) permeability

(iii) cell as osmotic system. (6)

8. (a) Explain the factors affecting photosynthesis. (6)

(b) Write about the electron transport pathway in chloroplast membranes. (7)

(c) Describe light dependent phase of photosynthesis. (7)

DE–4157 **22**

DISTANCE EDUCATION

M.Sc. (Botany) DEGREE EXAMINATION, MAY 2012.

(Specialization plant Bio-technology)

CELL BIOLOGY, GENETICS AND PLANT BREEDING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Write an essay on structure and function of nucleus. (7)

   (b) Describe the structure of Golgi complex and its functions. (7)

   (c) Explain the organization and functions of cytosketons. (6)

2. (a) Enumerate the functions and assembly of biological membrane. (7)

   (b) Write notes on transport of ions and molecules across the membrane. (6)

   (c) Describe the protein sorting in ER. (7)

3. (a) Explain the gene interaction with suitable example. (7)

   (b) Give an account on sex linked inheritance. (7)

   (c) Write notes on male sterlity. (6)

4. (a) Write an essay on gene mapping. (6)

(b) Explain cytoplasmic inheritance. (7)

(c) Describe origin and induction of prions. (7)

5. (a) Write an essay on types of polyploidy. (7)

(b) Explain mutation and its types. (6)

(c) Write short notes on population genetics. (7)

6. (a) Write notes on genetic variability and its role in plant breedings. (7)

(b) Explain the vegetative propagation in plant breeding. (6)

(c) Describe breeding for disease resistance and stress tolerance. (7)

7. (a) Explain the structure and functions of chloroplast. (6)

(b) Write down the differences between Eukaryotic and prokaryotic cells. (7)

(c) Describe protein sorting in nucleus. (7)

8. (a) Write about the objectives of plant breeding. (7)

(b) Explain gene interaction. (7)

(c) Describe inbreeding depression theories. (6)

DISTANCE EDUCATION

M.Sc. (Botany) DEGREE EXAMINATION, MAY 2012.

Specialization Plant Bio-Technology

BIOTECHNIQUES IN BOTANY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carries 20 marks.

$(5 \times 20 = 100)$

1. (a) What is Agarose Gel electrophoresis. Explain its principle and applications. (8)

   (b) Short notes on SDS- PAGE. (6)

   (c) Explain Two dimensional (2D) electrophoresis. (6)

2. (a) Short notes on HPLC. (7)

   (b) Explain TEM. (7)

   (c) Explain the types of Centrifuge. (6)

3. (a) Short notes on Auto Radiography. (6)

   (b) Write the principle and application of Radio Active Isotopes. (8)

   (c) Write about Scintillation counter and its types. (6)

4. (a) Explain Blotting techniques. (10)

   (b) Short notes on AFLP. (5)

   (c) Write about Hybridisation (5)

5. (a) Difference between Transcriptomes and Proteomes. (7)

(b) What is Bioinformatics? Write its application in our field. (8)

(c) Write about Tracking gene expression. (5)

6. (a) Short notes on Ultracentrifuges. (8)

(b) Short notes on FPLC and its application. (6)

(c) Explain Confocal Microscopy. (6)

7. (a) Explain Native-PAGE. (6)

(b) Write the method of sample preparation for SEM. (8)

(c) Write the Sedimentation rate and Sedimentation Coefficient. (6)

8. (a) Explain the types of PCR. (10)

(b) Write short notes on genome. (5)

(c) Applications of DNA sequencing. (5)

DE–4159 **24**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization – Plant Biotechnology)
DEGREE EXAMINATION, MAY 2012.

BIODIVERSITY CONSERVATION AND IPR

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) Define Biodiversity and its types. (4)

   (b) Discuss the case study of Basmati rice. (8)

   (c) Define types of biodiversity conservation. (8)

2. (a) Describe biodiversity Hotspots in India. (8)

   (b) Define Endemism. (4)

   (c) Narrate Wildlife Sanctuaries and National Parks in India. (8)

3. (a) Discuss Threatened plants and animals in India. (8)

   (b) Discuss Biodiversity act of India 2002 and 2004. (8)

   (c) Comment on IPR. (4)

4. (a) Narrate the plant genetic resources in India. (4)

(b) Describe GATT and WTO and their role in Biodiversity conservation. (8)

(c) Describe Patenting life forms and their impact on Biodiversity. (8)

5. (a) Comment on Patent application. (8)

(b) Discuss GM food and give impacts on environment. (8)

(c) Define Trade mark, Trade Secret and Copy right. (4)

6. (a) Discuss plant breeder's right. (8)

(b) Briefly describe concepts of Biodiversity. (8)

(c) Discuss General guidelines for research in transgenic plants. (4)

7. (a) List out Good Laboratory Practices (GLP). (4)

(b) Discuss the Rules for patenting. (8)

(c) Write in detail about Red Data book. (8)

8. (a) Briefly discuss values and uses of biodiversity. (4)

(b) Discuss importance of Agro biodiversity. (8)

(c) Write short notes on phytogeographic zones. (8)

DE–4160 **25**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Biotechnology) DEGREE EXAMINATION, MAY 2012.

PLANT GENETIC ENGINEERING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the scope and importance of plant genetic engineering. (6)

   (b) Comment on plant genome project. (7)

   (c) Short notes on restriction enzymes. (7)

2. (a) Write notes on *rpt II* gene in plant transformation. (7)

   (b) Write about any three techniques which are very important in plant genetic engineering. (7)

   (c) Comment on *Ti* plasmid. (6)

3. (a) *Agrobacterium tumefaciens* : An important vector for genetic engineering. (7)

   (b) Write about gene transfer by microinjection technique. (7)

   (c) Describe PEG mediated transformation. (6)

4. (a) Role of protease inhibitors in insect control. (7)

   (b) Write about Bt — based genetic modification of plants. (7)

   (c) Comment on herbicide resistance in plants. (6)

5. (a) Briefly explain the advantages of chloroplast genetic engineering. (7)

(b) Write about plasmids. (6)

(c) Comment on plant genome project. (7)

6. (a) Write short notes on inducible promoters. (7)

(b) What are the features which are common to cloning vectors? (6)

(c) Describe the role of selectable markers in transformation with one or two example. (7)

7. (a) Write about Antisense RNA mediated protection. (7)

(b) Discuss about gene silencing. (8)

(c) Write noted on 'cry' genes. (5)

8. (a) Describe the biotechnological approaches to protect the plants against fungal pathogens. (6)

(b) Short notes on transposons. (6)

(c) Comment critically on genetically modified organisms (GMOs)-Promise and danger. (8)